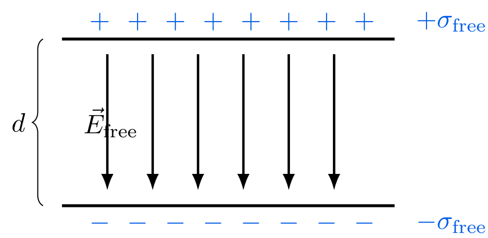
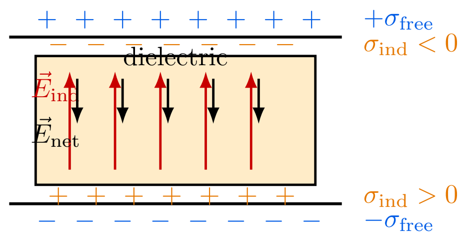
Revisiting the capacitor problem (dielectric reduces the field and hence the potential)

Assume two large parallel conducting plates (air/vacuum gap) separated by distance d. The plates carry **free** surface charge densities $\pm\sigma_{\text{free}}$. Initially, the potential difference is ϕ_0.



Step 1: no dielectric

Uniform field in the gap: $E_{\text{free}} = \dfrac{\sigma_{\text{free}}}{\varepsilon_0}$.



Step 2: insert dielectric

Induced charges create $\vec{E}_{\text{ind}}$ opposing the free-charge effect.

Figure 1: Parallel-plate capacitor revisited: introducing a dielectric creates induced (bound) charges and reduces the net field.

Step 1: Field with only free charges (no dielectric)

For two large plates with free surface charge densities $\pm\sigma_{\text{free}}$, the uniform field between them is

$$\boxed{E_{\text{free}} = \frac{\sigma_{\text{free}}}{\varepsilon_0}} \qquad \text{(inside the gap; outside fields cancel).}$$

Step 2: Introduce a dielectric $\Rightarrow$ induced charges and induced field

When a dielectric is inserted, its molecules polarize and produce **induced/bound** surface charge densities $\pm\sigma_{\text{ind}}$ on the dielectric faces. These induced charges create an induced field of magnitude

$$\boxed{E_{\text{ind}} = \frac{\sigma_{\text{ind}}}{\varepsilon_0}}$$

and its direction is opposite to the free-charge effect, so

$$\boxed{\vec{E}_{\text{net}} = \vec{E}_{\text{free}} + \vec{E}_{\text{ind}} \;\Rightarrow\; E_{\text{net}} = E_{\text{free}} - E_{\text{ind}}.}$$

Linear dielectric assumption (engineering range)

Experiments show that in the linear regime,

$$\sigma_{\text{ind}} = \alpha\,\sigma_{\text{free}} \quad \Rightarrow \quad E_{\text{ind}} = \alpha\,E_{\text{free}}, \qquad (0 < \alpha < 1),$$

hence

$$E_{\text{net}} = E_{\text{free}}(1-\alpha).$$

We define the dielectric constant (relative permittivity) ε_r by the observed reduction:

$$\boxed{E_{\text{net}} = \frac{E_{\text{free}}}{\varepsilon_r}}$$

Therefore,

$$\boxed{E_{\text{net}} = \frac{\sigma_{\text{free}}}{\varepsilon_0\,\varepsilon_r}}$$

From field to potential

Since the field in the gap is uniform, the potential difference is

$$\phi = -\int \vec{E}\cdot d\vec{l} = -E_{\text{net}}\,d.$$

If the free charge on the plates is unchanged (isolated capacitor), then

$$\boxed{\phi = \frac{\phi_0}{\varepsilon_r}} \qquad \text{(because } E_{\text{net}} = E_{\text{free}}/\varepsilon_r\text{)}.$$

Takeaway

$$E_{\text{free}} = \frac{\sigma_{\text{free}}}{\varepsilon_0}, \qquad E_{\text{net}} = E_{\text{free}} - E_{\text{ind}} = \frac{E_{\text{free}}}{\varepsilon_r} = \frac{\sigma_{\text{free}}}{\varepsilon_0\varepsilon_r}, \qquad \phi = E_{\text{net}}d = \frac{\phi_0}{\varepsilon_r}.$$

So introducing a dielectric reduces the field and hence reduces the potential difference (for fixed free charge).

What happens to ϕ when we change plate separation after disconnecting the supply?

Setup A parallel-plate capacitor (plate area A, separation d) is connected to a DC source and an ammeter in series. A voltmeter is connected across the capacitor.

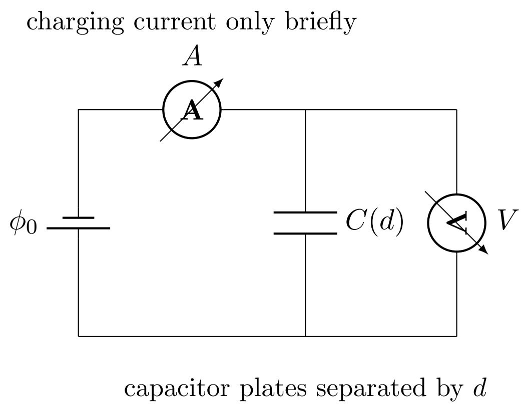
Step 1: Charge the capacitor (with $d = 1\,\text{mm}$, $\phi_0 = 10\,\text{V}$)

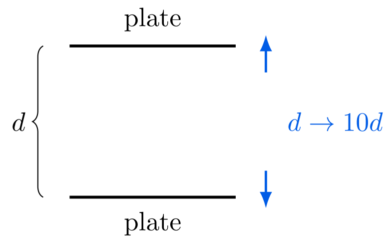
While the battery is connected, charge flows for a short time. The ammeter shows a transient current during charging. Once steady state is reached,

$$\phi(a \text{ across plates}) = \phi_0 = 10\,\text{V}.$$

The capacitance at separation d is

$$C_0 = \frac{\varepsilon A}{d},$$





(b) After disconnecting the source, increase separation.

(a) Circuit: ammeter in series, voltmeter across the capacitor.

Figure 2: Thought experiment used to distinguish **fixed voltage** vs **fixed charge** behavior of a capacitor.

so the **free charge** stored is

$$\boxed{Q_{\text{free}} = C_0\,\phi_0}$$

(on the two plates: $+Q_{\text{free}}$ and $-Q_{\text{free}}$).

Step 2: Disconnect the power supply, then increase separation to $10d$

Now the capacitor is isolated (no conducting path for charge to leave/enter the plates). Hence,

$$\boxed{Q_{\text{free}} = \text{constant}} \qquad \Rightarrow \qquad \boxed{\text{ammeter reads } 0}$$

because there is no charge flow through the series branch after disconnection.
But the capacitance changes when d changes:

$$C_1 = \frac{\varepsilon A}{10d} = \frac{C_0}{10}.$$

Since Q_{free} is unchanged,

$$\phi_1 = \frac{Q_{\text{free}}}{C_1} = \frac{C_0\phi_0}{C_0/10} = 10\,\phi_0.$$

Therefore,

$$\boxed{\phi_1 = 10\,\phi_0 = 100\,\text{V}}$$

and the voltmeter shows the increased voltage.

Check using the electric field

For an ideal parallel-plate capacitor (neglect fringing),

$$E = \frac{\sigma_{\text{free}}}{\varepsilon} = \frac{Q_{\text{free}}}{\varepsilon A}.$$

Since Q_{free} does not change, E **does not change**. But the potential difference is $\phi = Ed$, so if d increases by $10\times$, then ϕ increases by $10\times$.

Takeaway (exactly what this thought experiment teaches).

- While connected to the source: $\phi = \phi_0$ is fixed, current flows only during charging.
- After disconnecting the source: the capacitor is isolated $\Rightarrow Q_{\text{free}}$ stays constant $\Rightarrow$ ammeter reads zero.
- Increasing plate separation reduces capacitance: $C \propto 1/d$.
- With Q_{free} fixed, voltage increases: $\phi = Q/C \propto d$. For example if: $d \to 10d \Rightarrow \phi : 10\text{ V} \to 100\text{ V}$.

(Optional remark) Where did the extra electrical energy come from? With Q_{free} fixed, the stored energy is

$$U = \frac{Q_{\text{free}}^2}{2C}.$$

Since C decreases when d increases, U increases. This extra energy comes from the *mechanical work* done in pulling the plates apart against electrostatic attraction.

Insert a dielectric after disconnecting the supply (ammeter reads zero)

Where we are (from previous step). After charging to $\phi_0 = 10\,\text{V}$ at $d = 1\,\text{mm}$, we disconnected the supply and increased the plate spacing to

$$d' = 10d = 10\,\text{mm} \quad \Rightarrow \quad \phi' = 10\phi_0 = 100\,\text{V},$$

with **no charge flow** (ammeter reads zero) because the capacitor is isolated:

$$Q_{\text{free}} = \text{constant}.$$

Step 3: Insert a dielectric of relative permittivity ε_r while the capacitor is isolated

Now insert a dielectric slab completely filling the gap (same area A, same separation d'). Since the capacitor is still disconnected from the source, **free charge cannot change**:

$$\boxed{Q_{\text{free}} \text{ does not change } \Rightarrow \text{ ammeter reads } 0.}$$

However, the dielectric polarizes and creates induced (bound) charges, reducing the field in the medium.

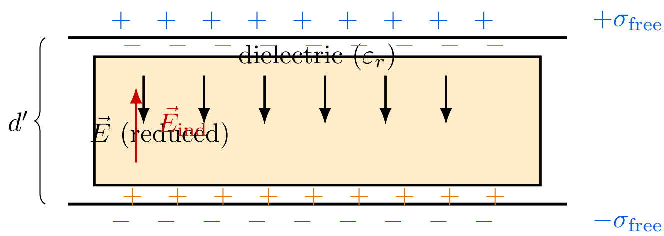


Figure 3: Insert dielectric after disconnecting the source: Q_{free} unchanged, but $\vec{E}$ and hence ϕ reduce by ε_r.

Field, potential, and capacitance

Because the dielectric reduces the field by the factor ε_r,

$$\boxed{E = \frac{\sigma_{\text{free}}}{\varepsilon_0 \varepsilon_r}} \tag{1}$$

(Here $\sigma_{\text{free}} = Q_{\text{free}}/A$ is unchanged.)
The potential difference is still

$$\boxed{\phi = E\, d'} \tag{2}$$

so inserting the dielectric reduces the voltage by ε_r:

$$\phi_{\text{new}} = \frac{1}{\varepsilon_r}\,\phi' \quad \Rightarrow \quad \boxed{\phi_{\text{new}} = \frac{\phi'}{\varepsilon_r}}$$

With the note's numbers: if $\phi' = 100\,\text{V}$, then $\phi_{\text{new}} = 100\,\text{V}/\varepsilon_r$.
Finally, capacitance is defined by

$$\boxed{C = \frac{Q_{\text{free}}}{\phi}} \tag{3}$$

Since Q_{free} is unchanged while ϕ reduces by ε_r, the capacitance increases by ε_r:

$$C_{\text{new}} = \frac{Q_{\text{free}}}{\phi_{\text{new}}} = \frac{Q_{\text{free}}}{\phi'/\varepsilon_r} = \varepsilon_r \frac{Q_{\text{free}}}{\phi'} = \varepsilon_r\, C'.$$

Equivalently (parallel plate):

$$\boxed{C_{\text{new}} = \frac{\varepsilon_0 \varepsilon_r A}{d'}}$$

Takeaway. For the isolated capacitor (Q_{free} fixed):

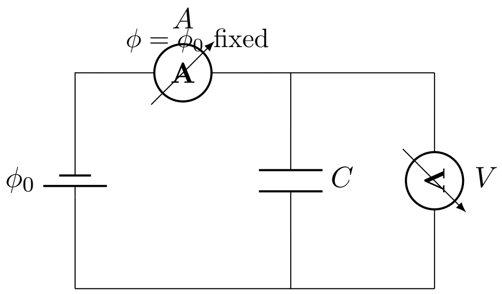
$$E = \frac{\sigma_{\text{free}}}{\varepsilon_0 \varepsilon_r}, \qquad \phi = E\, d', \qquad C = \frac{Q_{\text{free}}}{\phi} = \frac{\varepsilon_0 \varepsilon_r A}{d'}.$$

So inserting a dielectric causes $E \downarrow$ by ε_r, $\phi \downarrow$ by ε_r, and $C \uparrow$ by ε_r (with ammeter still reading zero).

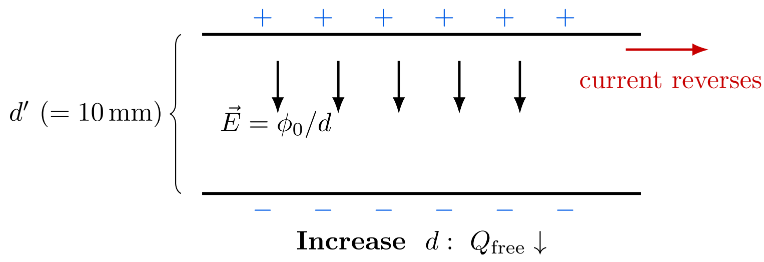
keep the power supply connected (so ϕ is fixed)

We repeat the same capacitor setup (voltmeter across, ammeter in series), **but this time we do not disconnect the battery**. Hence the battery *enforces* a fixed potential difference:

$$\boxed{\phi = \phi_0 = 10\,\mathrm{V} \text{ (fixed as long as the supply is connected).}}$$



(a) Battery connected $\Rightarrow \phi$ fixed.



(b) At fixed ϕ_0, increasing d makes Q_{free} decrease (charge flows away).

Figure 4: Battery connected: ϕ is fixed. Changing d or inserting dielectric changes Q_{free}.

(2) Increase plate separation while the battery is still connected

Start with $d = 1\,\mathrm{mm}$, $\phi = \phi_0 = 10\,\mathrm{V}$. Now increase separation to

$$d' = 10d = 10\,\mathrm{mm}, \qquad \text{but keep the battery connected.}$$

Because the battery fixes ϕ,

$$\boxed{\phi = \phi_0 \text{ does not change even when } d \text{ changes.}}$$

For an ideal parallel-plate capacitor (neglect fringing),

$$E = \frac{\phi}{d}.$$

Hence, when d increases to $d' = 10d$,

$$\boxed{E' = \frac{\phi_0}{d'} = \frac{\phi_0}{10d} = \frac{E_0}{10}} \quad \Rightarrow \quad \text{field reduces by } 10\times.$$

Capacitance becomes

$$C' = \frac{\varepsilon_0 A}{d'} = \frac{\varepsilon_0 A}{10d} = \frac{C_0}{10}.$$

But $Q_{\text{free}} = C\phi$ (battery keeps $\phi = \phi_0$), so

$$Q'_{\text{free}} = C'\phi_0 = \frac{C_0}{10}\phi_0 = \frac{Q_{\text{free},0}}{10}.$$

$$\boxed{Q_{\text{free}} \text{ must reduce by } 10\times \Rightarrow \text{charge flows away from the plates.}}$$

Therefore the ammeter shows a transient current in the *reverse* direction during the change.

(4) Now insert a dielectric (ε_r) but keep the battery connected

Keep d' fixed (say $d' = 10\,\text{mm}$) and still keep $\phi = \phi_0 = 10\,\text{V}$. Now insert a dielectric completely filling the gap with relative permittivity ε_r.
Capacitance increases:

$$\boxed{C_{\text{new}} = \frac{\varepsilon_0 \varepsilon_r A}{d'} = \varepsilon_r C'.}$$

Since the battery still fixes $\phi = \phi_0$, the free charge becomes

$$Q_{\text{free,new}} = C_{\text{new}}\phi_0 = \varepsilon_r C'\phi_0 = \varepsilon_r Q'_{\text{free}}.$$

$$\boxed{Q_{\text{free}} \text{ increases by } \varepsilon_r \Rightarrow \text{charge flows into the plates.}}$$

So the ammeter shows a transient current *into* the capacitor plates.

Important statement Because ϕ and d' are both fixed here,

$$\boxed{E = \frac{\phi_0}{d'} \text{ does not change after inserting the dielectric.}}$$

The dielectric polarization *tends* to reduce the field (via induced charges), but the battery supplies extra free charge so that the *net* field remains ϕ_0/d'. That is why Q_{free} must increase.

Takeaway: all four scenarios (with and without power supply)

Key rule:

- **Battery connected** $\Rightarrow$ $\boxed{\phi \text{ is fixed}}$ (so $Q_{\text{free}} = C\phi$ adjusts).
- **Battery disconnected (isolated capacitor)** $\Rightarrow$ $\boxed{Q_{\text{free}} \text{ is fixed}}$ (so $\phi = Q_{\text{free}}/C$ adjusts).

Let initial state be: d, air gap, ϕ_0, $C_0 = \varepsilon_0 A/d$, $Q_0 = C_0\phi_0$. Consider changing separation to $d' = k\,d$ (e.g. $k = 10$) and/or inserting dielectric ε_r fully.

1. **(Battery disconnected) Increase separation $d \to d' = kd$:**

$$Q_{\text{free}} = \text{const} = Q_0, \quad C = \frac{\varepsilon_0 A}{d'} = \frac{C_0}{k}, \quad \boxed{\phi = \frac{Q_0}{C} = k\,\phi_0}$$

Ammeter reads **zero** (no charge flow after disconnection). Field: $E = \sigma_{\text{free}}/\varepsilon_0$ **unchanged** (since Q_{free} unchanged), so $\phi = Ed$ increases with d.

2. **(Battery disconnected) Insert dielectric (ε_r), with d' fixed:**

$$Q_{\text{free}} = \text{const}, \quad C_{\text{new}} = \frac{\varepsilon_0 \varepsilon_r A}{d'} = \varepsilon_r C, \quad \boxed{\phi_{\text{new}} = \frac{\phi}{\varepsilon_r}}$$

Ammeter reads **zero**. Field reduces: $\boxed{\mathrm{E}_{\text{new}} = E/\varepsilon_r}$.

3. **(Battery connected) Increase separation $d \to d' = kd$:**

$$\boxed{\phi = \phi_0 \text{ (fixed)}}, \quad C = \frac{C_0}{k}, \quad \boxed{Q_{\text{free}} = C\phi_0 = \frac{Q_0}{k}}$$

Charge must **leave** the plates $\Rightarrow$ ammeter shows a transient current (reverse direction). Field reduces: $\boxed{\mathrm{E}{=}\phi_0/d' = E_0/k}$.

4. **(Battery connected) Insert dielectric (ε_r), with d' fixed:**

$$\boxed{\phi = \phi_0 \text{ (fixed)}}, \quad C_{\text{new}} = \varepsilon_r C, \quad \boxed{Q_{\text{free,new}} = \varepsilon_r Q_{\text{free}}}$$

Charge must **enter** the plates $\Rightarrow$ ammeter shows a transient current into the capacitor. **Important:** with ϕ and d' fixed, $\boxed{\mathrm{E}{=}\phi_0/d' \text{ does not change}}$; dielectric polarization is compensated by extra free charge supplied by the battery.